Week 1 — Introduction to Blockchain

1) Why the lecture starts with Bitcoin statistics

The lecture opens with Bitcoin numbers: around **19.98 million bitcoins in circulation**, a **maximum supply of 21 million**, about **1.02 million left to be mined**, about **144 blocks per day**, around **450 new bitcoins per day** at the current reward, a blockchain size of about **715 GB**, and only about **7 transactions per second**, compared to Visa's roughly **24,000 transactions per second**. It also mentions that around **4 million bitcoins may be lost forever**.

Why is sir starting with this instead of first giving a definition? Because he wants to show two things at once.

First, Bitcoin is not some tiny experiment. It is a huge real-world system with massive value, millions of users, a giant chain, and a long-running history. So blockchain is not just a theory topic. It exists in practice at very large scale. The fact that the chain is hundreds of gigabytes also tells you that blockchain data accumulates over time and can become very large.

Second, these numbers also quietly show Bitcoin's limitations. The lecture specifically points out that Bitcoin does around **7 transactions per second**, while Visa does roughly **24k**. That comparison is not random. It is teaching you that a blockchain can be revolutionary, but it is not automatically faster or better than all traditional systems in every metric. Blockchain often gains things like decentralization and tamper-resistance, but it may lose on speed, throughput, and storage efficiency.

Third, the fixed supply of **21 million** matters because it introduces one early property of Bitcoin: scarcity. Unlike normal fiat currency, which governments can issue more of, Bitcoin has a built-in upper limit. Even if this week is not yet the "money" lecture, sir is already planting the idea that blockchains can encode rules directly into the system. That is a very important mindset for later topics.

So this opening slide is really doing a lot:
Bitcoin is huge, Bitcoin is limited, and Bitcoin is only one example.

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2) The cryptocurrency avalanche — why this matters

Then the lecture talks about the explosion of cryptocurrencies: **18,000+ active cryptocurrencies**, hundreds on Coinbase and Binance, **500–600+ million users**, **1000+**

exchanges, and many businesses accepting crypto. But immediately after that, the slides say: **we will not cover crypto for the reasons they are famous for**. Instead, the focus is **blockchain technology**, and more broadly **distributed ledger technology (DLT)**, whose concepts are older than crypto hype.

This is a very important conceptual reset.

Many people hear the word "blockchain" and think:
price,
trading,
coins,
profit,
scams,
investment,
or market crashes.

Your sir is deliberately separating the **technology** from the **hype**.

The point is:

- Crypto is one application.
- Blockchain is the underlying technology.
- DLT is the even broader family of ideas behind shared distributed records.

So if a short question asks, "Why does blockchain not always mean Bitcoin?" your answer should not just be "because there are many cryptocurrencies." That is too shallow. The better explanation is: Bitcoin is only one use case of blockchain, and blockchain itself is part of the wider idea of distributed ledgers. The course is about the mechanisms and architecture, not just about cryptocurrency fame.

Also notice the slide says blockchain concepts are around for about **30 years** and DLT ideas for about **40 years**. That means the course is telling you that the intellectual roots of this field existed before Bitcoin became famous. Bitcoin made the idea famous, but did not invent every underlying concept from nothing.

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3) What this course is teaching — and what it is not

The course slide says it will teach:
the underlying technologies to construct a blockchain, the properties and pitfalls of blockchains, smart contracts, cryptocurrencies as a use case, and financial applications such as stablecoins and tokenization. It will **not** teach how to open a crypto account, investment strategies, or regulatory prescriptions.

This matters for your midterm because it tells you what kind of answers sir probably expects.

He wants technical and conceptual understanding:

- What is a block?
- What is a hash?
- Why does consensus matter?
- What makes a chain valid?
- What are the benefits and tradeoffs?

He does **not** want stock-market style answers like:
“blockchain is good because price goes up.”

Even if the question sounds broad, answer in a systems way, not in an investor way.

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4) The motivation for creating blockchain

This is one of the most important parts of Week 1.

The lecture says the growing need for digital payments led to several problems: intermediaries, double-spending, missing financial freedom, and complicated global transactions. It also says the need was for **direct P2P payments**, and that Satoshi Nakamoto created blockchain technology in 2009.

Let’s unpack every single part carefully.

4.1 Intermediaries

An intermediary is a middleman. In payments, that middleman is usually a bank, payment processor, wallet company, exchange, or another central service.

Why is this a problem?

Because when a middleman is required, the payment depends on that middleman’s decisions and availability. The lecture says intermediaries can:

- prohibit payment,
- reverse payment,
- and may not always be online.

That means if A wants to pay B, the payment is not purely between A and B. A third party must approve or process it. That third party may:

- block it,

- delay it,
- reverse it,
- freeze it,
- charge extra,
- or simply be down.

So one deep motivation of blockchain is: can two parties transfer value directly without a trusted central middleman?

That is what “peer-to-peer” really means here. It means the system should work **between participants directly**, not only through a central controller.

4.2 Double-spending

This is a foundational blockchain problem.

The slide asks: **Can you double-spend physical currency notes?**

Think about physical cash. If I hand you a real Rs. 1000 note, I do not still physically have that same note. So spending the exact same note twice is hard in normal circumstances.

But digital data behaves differently. Digital information can be copied. If money is represented digitally, what stops me from copying the same payment information and sending it to multiple people?

That is the double-spending problem:
the risk that the same digital unit of value is spent more than once.

The slide says this is an inherent problem in payments that are both **P2P and digital**. That phrase is very important. If payments are digital and there is no central bank or payment server to keep the master record, then how do all participants agree that a coin was spent only once? That is one of the core problems blockchain tries to solve.

4.3 Missing financial freedom

The slide mentions **the bailouts of 2008**.

You do not need a long economics essay here, but the idea is that central financial systems made many people feel uncomfortable about dependence on traditional institutions. The blockchain vision included a desire for more financial autonomy — a system where rules are enforced by protocol and network consensus rather than purely by trusted institutions.

So when sir says “missing financial freedom,” it means people wanted a system in which:

- value transfer is less dependent on banks,
- access is less controlled by centralized institutions,

- and the system's rules are transparent and embedded in the protocol.

4.4 Complicated global transactions

The lecture says global transactions need to be **fast and cheap**.

Traditional international payments can be:

- slow,
- expensive,
- dependent on multiple intermediaries,
- and limited by banking hours or cross-border infrastructure.

Blockchain was motivated partly by the hope of making digital value transfer more direct across borders.

4.5 The big picture motivation

Put together, the motivation is:

People needed a way to make digital peer-to-peer payments where:

- no central intermediary is mandatory,
- double-spending is prevented,
- transfers can be more autonomous,
- and the system can support global digital transactions more cleanly.

That is the real "why blockchain?" of Week 1.

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5) What exactly is blockchain?

The slide defines blockchain as **a ledger for recording data, just like databases**, used to record transactions across a network of computers called **nodes**. It also says transactions may be initiated by any node or by an end user/organization on their behalf, and that most nodes keep part of the ledger while some keep a full ledger.

Now let's explain this slowly.

A **ledger** is basically a record book. Think of a register or account book where entries are written down. In older accounting, a ledger records who paid whom, who owes what, and what transactions happened.

So blockchain is not magical at first glance. At its heart, it is just a way of keeping records.

But the important twist is that it is **distributed**.

That means the record is not stored only in one place, on one company's server, under one administrator's control. Instead, many computers in the network hold and maintain versions of that ledger.

Those computers are called **nodes**.

A node is simply a machine running the blockchain software. Depending on the design, a node may:

- store data,
- validate data,
- relay data,
- or help maintain the chain.

The slide also says some nodes keep only **part** of the ledger, while some keep the **full** ledger. That means not every participant must always store the entire historical record. Some do; some do not. This matters later for scalability and verification.

Also, who creates transactions? The lecture says:

- any node in the blockchain network,
- or an end user / organization on its behalf.

So if I, as a user, send a payment, I might do it through software or a service like an exchange, but the transaction still enters the blockchain network through nodes.

A good exam definition would be:

Blockchain is a distributed ledger system where transaction data is recorded across a network of nodes rather than being maintained by a single central authority.

But for understanding, remember it like this:
a shared record book spread across many computers.

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6) Why is it called a blockchain?

The lecture says it is called a blockchain because of its structure:
transactions are written in **blocks** of data, and those blocks are chained together through **cryptographic hashing**.

This part sounds simple, but it is very exam-worthy.

6.1 What is a block?

A block is a container of data. In this lecture, that data is mainly transactions.

So instead of writing each transaction as a completely separate independent item in a giant loose list, blockchain groups a set of transactions into one block.

For example:
Block 55 may contain transactions 1 to 76,
Block 56 may contain another bunch,
Block 57 another bunch, and so on.

6.2 What makes it a chain?

It becomes a chain because each block is linked to the previous block using hash information.

So the structure is not:
Block 1, Block 2, Block 3 as separate boxes.

It is:
Block 2 points back to Block 1,
Block 3 points back to Block 2,
and so on.

That linking is why it is a chain of blocks, not just a pile of blocks.

6.3 Why this matters

This chained structure is what gives blockchain much of its tamper-evident property. If blocks were not linked, changing one old block might not affect later blocks. But because they are linked, old changes ripple forward.

That idea becomes clearer once we understand hashing.

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7) Cryptographic hashing — the heart of the chain

The lecture says a cryptographic hash function:

- maps input text to a fixed-length, random-looking output,
- can be computed by known software such as SHA-256,
- and any change in input text causes a significant change in the hash output.

This is one of the most important concepts in all of blockchain.

7.1 What is a hash, in plain words?

A hash is like a digital fingerprint of data.

You give some input to a hash function:
maybe one word,
maybe a whole file,
maybe a block of transactions.

The function produces a short fixed-size output that represents that input.

The output is not a summary in normal English. It looks random and meaningless, like:
`A94D539C1` in the slide example.

7.2 Fixed-length output

This is important. The input can be tiny or huge, but the output length stays fixed.

So whether the block has small data or large data, the hash output remains the same size format.

That makes it efficient to use as an identifier or integrity check.

7.3 Random-looking output

The hash output looks random. That is useful because it becomes hard to predict the exact output you will get from a given input without actually computing it.

7.4 Tiny change, huge hash difference

This is one of the most important properties.

If the input changes even slightly — even a single character — the hash changes a lot.

The slide even shows the idea with changed text input producing a very different SHA-256 value.

Why is this important?
Because if someone edits block data, that edit is visible through the changed hash.

This is the basis of tamper detection.

7.5 Hashing is not encryption

The lecture does not explicitly compare these yet, but you should know not to confuse them.

Hashing is not mainly about hiding data. It is mainly about producing a fixed fingerprint from data so integrity can be checked.

In Week 1, the role of hashing is:

- detect change,
- link blocks,
- support mining rules.

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8) How blocks are chained using hashes

The lecture says each block contains the **hash (digest) of the previous block**.

This is the exact mechanism of chaining.

Let’s imagine:

- Block 55 has some transactions.
- You compute Block 55’s hash.
- That hash is written inside Block 56 as the “previous hash.”
- Then Block 56 has its own transactions and its own resulting hash.
- That hash gets written into Block 57.

So every block is carrying a pointer to the previous block, but that pointer is not just a block number. It is a **hash-based reference**.

This matters because a hash depends on the contents of the previous block. So if someone changes the previous block, its hash changes, and the reference no longer matches.

In simple words:
each block says,
“I expect the previous block to look exactly like this.”

If the previous block no longer looks like that, the chain breaks.

That is a brilliant design idea, and it is one of the most testable concepts in early blockchain lectures.

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9) Practical hashing and the need for criteria

Now the lecture adds another layer. It says that in a practical blockchain, hashes need to meet some criteria — for example, a hash must start with **3 zeros**. If the hash does not meet the criteria, the block is not allowed. Since transaction data is fixed, the block creator includes a fictitious number called a **NONCE** and keeps changing it until the criteria are met. This process is called **mining**.

This is a huge concept, so let's break it carefully.

9.1 Why impose a hash condition at all?

Why not accept any hash?

Because if block creation were too easy, anyone could quickly create or rewrite blocks. The system wants block creation to require computational effort in a Proof-of-Work setting.

So the network says:
"A valid block is not just any block. Its hash must satisfy a rule."

A simple teaching example is:
the hash must begin with 000.

Not every block will naturally do that. In fact, most will not.

9.2 But if transactions are fixed, what can be changed?

The transactions inside the block are normally fixed — they are the actual payment records and other data that the block is meant to contain.

If you cannot change the transactions just to get a nicer hash, what do you change?

You add a **NONCE**.

9.3 What is a nonce?

A nonce is an extra value included in the block that can be changed freely.

It does not represent a real payment. It is there mainly to affect the block's hash output.

The miner tries:

- nonce = 1,
- compute hash,
- did it start with 000? no.

Then:

- nonce = 2,
- compute hash,
- still no.

Again and again and again.

Eventually, one nonce produces a hash that meets the required pattern. Then the block is considered valid under that rule.

9.4 What is mining?

Mining is this repeated trial-and-error process of cycling through nonce values until a valid hash is found. The recap slide literally says mining means to cycle through the nonce to meet some hash criteria.

So when someone says “miners are solving puzzles,” what they really mean in simple terms is: miners are repeatedly changing the nonce and hashing the block until the output satisfies the difficulty rule.

This is why Proof of Work consumes energy. It is computational work.

9.5 Why this is important for security

If it takes serious effort to produce a valid block, then rewriting history also becomes expensive. This helps make the chain harder to tamper with.

So hashing is not only used for linking blocks. It is also used as part of the cost of block creation.

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10) Recap of a typical block’s contents

The lecture recap says each block in a typical blockchain contains:

1. transactions,
2. hash of the previous block,
3. nonce.

This simple three-part structure is extremely important for exams.

Transactions

These are the actual records, like payments or transfers.

Previous block hash

This connects the current block to the earlier block, creating the chain.

Nonce

This is the variable number used during mining so that the current block's hash can satisfy the required criteria.

Even if sir gives a diagram question like "draw and label a block," this is the answer.

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11) What blockchain brings to the table

The lecture gives three big benefits:
immutability, **auditability**, and **decentralization**. It also briefly raises the questions of **privacy** and **anonymity**.

Let's study each deeply.

11.1 Immutability / tamper-proof

Immutability means that once data is written, it is extremely hard to change or roll back.

Notice the lecture wording: **nearly impossible**, not magically impossible. That wording matters.

Why is it hard to change?
Because if you change a transaction in a block, you change the contents of that block.
If contents change, the hash changes.
If the hash changes, the next block's stored previous hash no longer matches.
So you would need to recompute not just one block, but this block **and all subsequent blocks**.
The slide says this explicitly.

And in a PoW system, recomputing valid hashes is expensive because each block must satisfy the hash criteria again.

So immutability does not mean "nobody can ever change anything under any circumstances."
It means the system is structured to make unauthorized history changes extremely difficult and costly.

11.2 Auditability

Auditability means that transactions can be tracked and verified.

The lecture says:

- all transactions can be tracked and verified,
- all historical transactions are around and available.

This gives the blockchain a kind of traceable history. In a traditional private database, access to old records might be hidden or controlled. In a blockchain, transaction history is part of the system's design.

This is especially useful where multiple parties need a reliable trail of what happened.

11.3 Decentralization

The lecture says decentralization enables autonomy and censorship resistance, allows transactions without a central authority like banks, and reduces single points of failure.

This means no one central actor is supposed to have absolute control over whether the ledger exists, whether transactions are recognized, or whether the system continues to run.

Benefits include:

- less dependency on one institution,
- harder for one party to censor everyone,
- lower risk that one server failure takes down the whole system.

11.4 Privacy? Anonymity?

The slide simply raises these as questions.

That is important. It means:
do not casually assume that decentralization automatically means privacy.
A system can be decentralized but still not fully private.
A system can be auditable and transparent, which may reduce privacy.
So your sir is already hinting that these ideas are separate and more nuanced.

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12) What happens when a block is tampered with?

The lecture gives an example where a transaction changes from **Asim → Arif: $20** to **Asim → Arif: $50**, and then the hash changes and the blocks become "unchained." The slide says any change in the block data can be easily detected.

This is a core exam concept.

Let’s go step by step.

Suppose Block 55 contains the transaction “Asim → Arif: $20.”

Now an attacker edits that transaction to “Asim → Arif: $50.”

That tiny edit changes the contents of Block 55.

Since hash depends on contents, Block 55’s hash becomes different.

But Block 56 still stores the **old** expected previous hash of Block 55.

So Block 56 is now pointing to a version of Block 55 that no longer exists.

That mismatch breaks the chain linkage.

And if you now try to fix Block 56 by updating its previous hash, then Block 56’s own contents change, so Block 56’s hash changes too, which affects Block 57.

So one edit creates a domino effect forward.

That is why old tampering is easy to detect and expensive to repair.

This is one of the best “story” explanations to memorize:
change one old transaction, and every following block is disturbed.

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13) Can blockchain be permanently tampered with?

The lecture answers this carefully:
possible, but it would require **more than 50% of control**.

This is basically introducing the idea of a majority attack.

It means blockchain security is not absolute magic. It depends on the honest majority assumption. If an attacker controls most of the relevant network power or influence, then they may be able to outcompete or override the honest participants and push an alternative history.

For Week 1, you do not need to go beyond what the slide says. Just understand the message:

- Blockchain is strongly tamper-resistant.
- But not unbreakable in theory.
- Its security depends on distribution of control.

So if sir asks, “Is blockchain completely impossible to modify?” do **not** say yes. Say it is designed to be very hard to tamper with, and permanent tampering would typically require majority control.

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14) Security of financial transactions

The lecture asks:

- Has the payer authorized the transaction?
- Is the payee the right person?
- Can we verify transactions before writing them in a block?
- Is someone trying to spend more than what they have?
- Can this verification be done efficiently?

It mentions **digital signatures**, **public key cryptography**, and **simplified payment verification**.

This part is important because it reminds you that protecting block structure is not enough. We must also verify the **transactions themselves**.

Suppose Alice says she sends 0.3 BTC to Bob.

The blockchain must ask:

- Did Alice actually own 0.3 BTC?
- Did Alice really authorize this payment?
- Is Bob the intended receiver?
- Is this payment being duplicated?

So there are two layers of correctness:

1. Is the block structure valid?
2. Are the transactions inside it valid?

This lecture only introduces those questions. It does not fully solve them yet. Later topics like digital signatures and public/private keys answer the “Did Alice agree?” part.

For Week 1, the important lesson is:
a blockchain is not just about storing transactions — it must also validate them.

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15) When should blockchain be used?

The lecture says blockchain is useful when there is a large number of transactions that:

- are generated and possibly consumed by multiple parties,
- have a common interest,
- and need to avoid central control.

It says this boils down to a **multi-party consensus system**.

This is a very important practical question.

Blockchain is **not** for every problem.

If one trusted university office can maintain student attendance in one internal system, a normal database may be enough.

Blockchain becomes relevant when multiple different parties need to agree on shared data but do not want one party to be the sole controller.

Examples from the lecture’s style:

- everyone agrees that Muneeb transferred PKR 70k to Zain,
- everyone agrees a restaurant rating is 3.1/5.

Notice the second example is not even about money. That is important. Blockchain is about agreement on shared data, not only payments.

So the real use case test is:
Do multiple parties need a shared trusted record without relying fully on one central owner?

If yes, blockchain may make sense.

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16) The island/tribe example — why sir mentions it

The slide says: think of a tribe on an island a couple of thousand years ago. How can you do financial transactions without a bank?

This is actually a brilliant way to think.

The point is: banking is not the same thing as transaction recording. Human societies could still exchange value and keep track of who gave what to whom, even without modern banks.

So what matters is not “must there always be a bank?”
What matters is “how does the community agree on the record?”

Blockchain is a modern digital answer to that same ancient question:
How do many people agree on shared transaction history?

That is why the next topic is consensus.

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17) Consensus — why it is needed

The lecture says consensus is needed to develop a ledger, meaning a **single converged view** of the blockchain.

This is maybe the deepest idea of Week 1.

Imagine many nodes in a network.
One node hears about transaction A first.
Another node hears about transaction B first.
Some nodes are slow.
Some are offline.
Some may even be dishonest.

How do they all end up with one accepted history?

That process is consensus.

Consensus means:
the network participants converge on one shared version of the record.

Without consensus:

- one node may think Alice paid Bob,
- another may think she did not,
- another may think she paid someone else first.

Then there is no reliable ledger.

Consensus is what turns many local views into one accepted global view.

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18) Proof of Work and Proof of Stake

The lecture introduces two consensus options.

Proof of Work (PoW)

Block creators spend energy to create blocks, get rewards, and are called **miners**.

Absolutely — continuing from **Proof of Work and Proof of Stake**, and I'll keep the same detailed, study-from-this-alone style.

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18) Proof of Work and Proof of Stake

The lecture introduces two major ways a blockchain can reach consensus: **Proof of Work (PoW)** and **Proof of Stake (PoS)**. It says consensus is needed to develop a **single converged view of the blockchain**. In PoW, block creators spend energy and are called **miners**; in PoS, block creators stake cryptocurrency and are called **validators**. Bitcoin is shown as the example of PoW, while Ethereum is shown as the example of PoS.

Let's really understand this.

18.1 Why do we even need a rule for "who creates the next block"?

Imagine a blockchain network with thousands of nodes. All of them can see transactions. But if a new block has to be added, who gets to create it?

If everyone could freely add blocks whenever they wanted, the network would become chaotic:

- different people would add different versions,
- fake records could enter easily,
- and there would be no stable agreed chain.

So the blockchain needs a method to decide:

- who can propose the next block,
- why that block should be trusted,
- and how everyone else should accept it.

That method is the consensus mechanism.

18.2 Proof of Work (PoW)

In PoW, the right to create a block is tied to computational effort.

The lecture says block creators "purchase and spend energy to create blocks," and in return they receive some reward. These block creators are called **miners**.

What does "spend energy" mean in plain words?

It means miners use machines to keep computing hashes over and over while changing the nonce, until one of them finds a valid hash that satisfies the required condition. That repeated computation uses electricity and hardware resources.

So a PoW system says:
“If you want the right to add a block, prove that you did the computational work.”

That work is expensive. And because it is expensive, it becomes difficult for attackers to cheaply rewrite the chain.

This is why PoW is deeply linked to mining.

18.3 Proof of Stake (PoS)

In PoS, the right to participate in block creation is tied to economic stake rather than heavy computation.

The lecture says block creators “purchase and stake cryptocurrency,” receive a fee, and are called **validators**.

So instead of saying:
“Show me you burned energy,”

PoS says:
“Show me you have committed value into the system.”

That staked cryptocurrency acts like a form of commitment or skin in the game.

18.4 Main conceptual difference between PoW and PoS

You should remember the difference like this:

- **PoW**: security through computational work and energy expenditure
- **PoS**: security through economic stake and locked value

PoW makes it hard to attack by requiring large computing resources.
PoS makes it hard to attack by requiring large economic commitment.

18.5 Why this matters in Week 1

At this stage, the lecture is not going deep into all the technical details of PoS. The main goal is to make you understand that consensus is not one single universal thing. Different blockchains solve the “who gets to add the next block?” problem differently.

So if sir asks a short question like “What is the purpose of PoW/PoS?” the clean answer is:
they are mechanisms for achieving consensus and maintaining one agreed blockchain history.

19) When is a blockchain valid?

The lecture asks: **When is a blockchain legit?** And it answers: **when all the blocks are valid**. It then says a block is valid if:

- the correct hash of the previous block is present,
- the hash of this block meets the criteria in a PoW blockchain,
- and all transactions inside the block are valid.

This is extremely important because it teaches you that blockchain validity has layers.

19.1 A blockchain is not valid just because it exists

Just because someone gives you a sequence of blocks does not mean it is the legitimate blockchain.

A valid blockchain is one where **every block satisfies the rules**.

If even one block is incorrect, the entire chain's legitimacy becomes questionable.

19.2 Rule 1: Correct previous hash must be present

This ensures that the blocks are properly linked.

If Block 56 claims that it comes after Block 55, then the "previous hash" field inside Block 56 must actually match the real hash of Block 55.

If it does not match, the chain is broken.

19.3 Rule 2: The current block's hash must meet the required criteria

In a PoW system, it is not enough for the hash to merely exist. It must also satisfy the difficulty rule, such as beginning with the required number of zeros or fitting the current difficulty target.

This proves that the creator of the block did the required work.

19.4 Rule 3: Transactions inside the block must be valid

This is the most overlooked part.

Even if a block is perfectly linked and perfectly mined, it still cannot be accepted if it contains invalid transactions.

So a blockchain is not only about structure. It is also about transaction correctness.

A very good exam sentence is:

A block must be structurally valid and transactionally valid.

That sounds strong and mature.

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20) What makes a transaction valid?

The lecture gives the example: **Alice gives 0.3 BTC to Bob** and then asks:

- Did Alice have the currency?
- Did Alice agree?
- How do we indicate that Alice agreed?

This is another deeply important concept.

A blockchain transaction is not valid just because someone typed:
“Alice pays Bob.”

The system must verify real conditions.

20.1 Did Alice have the currency?

This means:
Does Alice actually own enough balance or spendable funds?

If Alice has only 0.1 BTC, she cannot validly send 0.3 BTC.

So the ledger must be checked against prior records.

20.2 Did Alice agree?

This means:
Was the transaction genuinely authorized by Alice, or is someone pretending to be Alice?

This is where later concepts like digital signatures and public key cryptography come in.

20.3 How do we indicate that Alice agreed?

This is asking:
How does the system see proof of authorization?

In normal banking, the bank may check your password, OTP, account login, or internal rules.

In blockchain, this is usually handled cryptographically.

For Week 1, the important takeaway is not the detailed mechanism yet, but the principle:

Transaction validity depends on ownership plus authorization.

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21) Process of blockchain formation

The slide has a heading called **Process: Blockchain Formation**. Even if the visual is not fully available in the parsed text, by combining the surrounding slides, we can explain what this process means in the Week 1 context. The block structure, transaction sharing, mining, validation, and convergence to a shared ledger are all parts of that formation process.

Here is the full story version of blockchain formation.

Step 1: Transactions are generated

Users or entities initiate transactions. These transactions enter the network through nodes. The lecture explicitly says transactions may be generated by any node, or by an end user or organization on their behalf.

Step 2: Transactions are shared across the network

Because the system is distributed, transaction information spreads among nodes rather than going to only one central server.

Step 3: Transactions are grouped into a block

A block creator takes a set of candidate transactions and packages them together in a block.

Step 4: The block references the previous block

The new block includes the hash of the previous block, so it can join the existing chain.

Step 5: The creator must satisfy the consensus rule

In PoW, the creator keeps changing the nonce until the block's hash satisfies the difficulty condition. This is mining.

Step 6: Other nodes can verify the block

The network checks:

- does the previous hash match?
- does the hash satisfy the rule?
- are the transactions valid?

Step 7: The block gets accepted into the growing chain

Once the network accepts it, the chain is extended.

Step 8: The updated history spreads and becomes part of the shared view

More and more nodes synchronize around this accepted version.

That, in plain words, is blockchain formation:
transactions are created, packaged, validated, linked, and accepted into a shared historical record.

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22) Centralized database vs distributed ledger

The lecture has a comparison slide called **Centralized Database vs Distributed Ledger**. Even though the parsed text mostly gives the heading and legend, the very next slides explain the core idea through journals and nodes, so we can understand the intended comparison well.

This is one of the most testable conceptual differences in Week 1.

22.1 What is a centralized database?

A centralized database is usually controlled by one central authority or organization.

For example:

- a bank's internal database,
- a university's student database,
- a company payroll system.

In such a system:

- one main party owns the infrastructure,
- one main party decides what updates are correct,
- users trust that central party to maintain the truth.

This is not automatically bad. In fact, centralized databases are often fast, efficient, and easier to manage.

22.2 What is a distributed ledger?

A distributed ledger is not owned or updated by just one central controller.

Instead:

- multiple nodes keep records,
- multiple participants share data,
- agreement has to emerge from synchronization and consensus.

This is more complex but reduces total dependence on one trusted center.

22.3 Main difference in one sentence

A centralized database relies on **central trust**.
A distributed ledger tries to create **shared trust without one central controller**.

22.4 Why this distinction matters

This is not only a technical distinction. It changes the entire architecture of the system.

In a centralized database:

- updates are usually fast because one authority decides.
- rollbacks and corrections are centrally controlled.
- failure or compromise of the central authority can affect the whole system.

In a distributed ledger:

- coordination is harder,
- consensus is necessary,
- but no single party is supposed to be the sole source of truth.

This is exactly why blockchains are powerful but also complex.

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23) DLT: the “journal” way of understanding it

The lecture gives one of the most important conceptual explanations of DLT:
Consider four nodes in a network. Transaction data is generated by them and shared. Each node builds a **journal**, meaning its own version of the database. For the example, there are four

journals, and the lecture asks: journals could be different — why? It then says: **the blockchain is the unified view of individual journals**.

This is one of the best ideas in the whole week, and many students miss it.

23.1 What is a journal here?

A journal is basically a local record or local copy maintained by a node.

Think of each node as keeping its own notebook.

Each notebook contains transaction history as that node currently sees it.

So in a four-node network:

- Node A has one journal
- Node B has one journal
- Node C has one journal
- Node D has one journal

23.2 Why might journals be different?

This is exactly the question the slide asks.

They might differ because:

- some nodes receive updates earlier than others,
- some nodes are temporarily offline,
- some transactions are still propagating,
- some blocks have not yet been accepted everywhere,
- network delays exist.

So distributed systems do **not** assume perfect instant global agreement.

This is a very mature point. Real systems have delays and partial views.

23.3 Then what is “the blockchain”?

This is the beautiful part.

The blockchain is not just “one database file sitting somewhere.”

It is the **unified view** that emerges when these individual journals synchronize closely enough and agree on the same accepted history.

So a very strong conceptual sentence is:

Blockchain is the agreed global view formed from many local journals.

That is much deeper than simply saying “blockchain is a database.”

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24) The goal of DLT

The next lecture slide says:

- different journals should be in sync most of the time,
- they should offer a shared view,
- syncing must happen without a central party,
- and the ledger is **not the database**; it is the **mechanism of syncing different journals**.

This is maybe the most subtle and brilliant statement in the lecture.

24.1 “In sync most of the time”

Notice the lecture does not say “perfectly identical every microsecond.”

It says “most of the time.” That reflects reality.

In a distributed network, temporary differences can exist. What matters is that the system converges and stays close enough to one shared accepted state.

24.2 “Offer a shared view”

The users of the blockchain need one practical answer to questions like:

- who owns what?
- which transactions happened?
- what is the current accepted record?

That is the shared view.

24.3 “Without a central party”

This is the core challenge. If a central party existed, syncing would be easy:
everyone just obeys the central server.

But blockchain wants that shared record **without** making one party the absolute controller.

That is why consensus becomes so important.

24.4 “The ledger is not the database”

This is the line you should memorize.

Why does the lecture say that?

Because students often imagine the ledger as if it were just a stored table.

But in this distributed setting, the ledger is better understood as:

- the agreed state,
- the synchronized outcome,
- the process/mechanism by which different journals converge.

So the ledger is not merely the physical storage. It is the shared agreed record produced through synchronization.

This is a very high-quality exam idea.

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25) Syncing: building consensus

The final Week 1 slide says:

- there is a set of journals,
- these are individual views of shared data,
- there must be convergence to a single shared view,
- that agreed view is the distributed ledger,
- the ledger is not quite a tangible thing but emerges from synchronization,
- and building distributed consensus is hard.

Let’s unpack that in plain English.

25.1 Set of journals = many local truths

Every node begins from its local perspective.

It knows what it has heard so far.
It knows what transactions or blocks it has seen.

But that does not yet guarantee that everyone sees exactly the same thing at the same time.

25.2 Convergence = coming together

Consensus is about convergence.

Convergence means that over time, the individual journals move toward one common accepted state.

This does not have to mean perfect instant uniformity from the very first second. It means the network agrees enough to maintain one accepted chain.

25.3 The distributed ledger emerges

The lecture says the ledger “just emerges from the efforts of synchronizing individual journals.”

This is a really powerful way of thinking.

The ledger is not some mystical object handed down from above.
It emerges from:

- message sharing,
- validation,
- block acceptance,
- and consensus.

25.4 Why is building distributed consensus hard?

Because many problems can happen:

- delays,
- different local views,
- failures,
- dishonest nodes,
- conflicting candidate blocks,
- no central authority to settle every dispute.

So Week 1 ends on a very important note:
the core challenge of blockchain is not simply storing data.
It is getting distributed participants to agree on shared data.

That is why consensus is at the heart of blockchain.

Week 1 complete story in one flow

Now let me give you the full story version in one connected narrative so it sticks in your head.

Human beings needed digital payments, but digital payments had real problems. Traditional systems relied on intermediaries like banks and payment processors. Those intermediaries could block, reverse, or delay payments, and they might not always be available. At the same time, digital money faced the double-spending problem: unlike physical cash, digital information can be copied, so the same digital value could be spent more than once if the system is not designed carefully. There was also a desire for greater financial autonomy and for global transactions that are faster and cheaper. This is the context in which blockchain was created.

Blockchain is basically a ledger, meaning a record of transactions, but unlike a normal centrally controlled database, it is spread across many computers called nodes. These nodes participate in maintaining the history of transactions. Some may store only part of the ledger, while some may store the full ledger. Transactions can be initiated by users, organizations, or nodes in the network.

It is called a blockchain because transactions are grouped into blocks, and those blocks are chained together. The chaining happens through cryptographic hashing. A hash function takes input data of any length and maps it to a fixed-length, random-looking output. If the input changes even a little, the output changes significantly. This makes hashes very useful for detecting tampering.

Each block contains the hash of the previous block. That means each block is tied to the exact contents of the one before it. If someone changes data in an earlier block, its hash changes, and the next block's reference to it no longer matches. This breaks the chain. That is why tampering is easy to detect.

In practical PoW blockchains, blocks are not accepted unless their hash satisfies some difficulty condition. Since transaction data is fixed, miners vary a nonce until they eventually find a valid hash. This repeated search is mining. Because it requires large computational effort, rewriting blocks is also expensive. That is one reason blockchain is tamper-resistant.

Blockchain therefore offers three major things: immutability, because changing old records is extremely hard; auditability, because historical transactions remain available and traceable; and decentralization, because transactions can be maintained without one central authority and with less single-point dependence. At the same time, the lecture hints that privacy and anonymity are separate questions and should not be casually assumed.

But secure block linking is not enough. The system must also verify that transactions themselves are valid. If Alice sends Bob 0.3 BTC, the system must know whether Alice really had the funds and whether she really authorized the payment. That is where later topics like digital signatures and public key cryptography come in.

Blockchain should be used when many parties generate shared data and need common agreement without depending on one central controller. In that case, the real problem becomes consensus: how can multiple participants arrive at one shared version of the truth? The lecture introduces Proof of Work and Proof of Stake as two major consensus approaches. PoW relies on energy expenditure and miners; PoS relies on staked value and validators. Both aim to create a single converged blockchain history.

Finally, the lecture reframes blockchain through DLT. Each node keeps its own journal, which is its local version of the shared data. These journals may temporarily differ. The blockchain is the unified agreed view that emerges when these journals synchronize. That is why the lecture says the ledger is not merely the database; it is the mechanism and result of syncing different journals into one distributed consensus. And that syncing problem is hard — which is exactly why blockchain is such an important systems topic.

What sir can test in detail from this week

From Week 1, sir can easily go deeper than definitions. He can ask things like:

Why is double-spending a problem only in digital P2P money and not in physical cash?
Why does changing one transaction force recomputation of later blocks?
Why is a hash useful if it does not “store” the original data?
Why is nonce needed if transactions are fixed?
Why is decentralization not the same as privacy?
Why is blockchain not suitable for every application?
Why is the ledger “not the database”?
Why can journals differ in DLT?
Why is consensus hard in a distributed environment?
Why does valid blockchain require both valid blocks and valid transactions?

Those are exactly the kinds of detailed short-answer questions that come from this lecture.

Ultra-short retention version

Memorize Week 1 like this:

Blockchain was created because digital payments had problems: intermediaries, double-spending, lack of autonomy, and slow expensive global transfers. It is a distributed ledger across nodes. Transactions are grouped into blocks. Each block contains transactions, the previous block’s hash, and often a nonce. Hashing gives a fixed-size fingerprint; tiny data changes create very different hashes. Each block stores the previous block’s hash, so tampering breaks the chain. In PoW systems, miners change the nonce until the hash satisfies the rule; this is mining. Blockchain brings immutability, auditability, and decentralization. A blockchain is valid only if each block is properly linked, satisfies consensus rules, and contains valid transactions. Blockchain is useful when many parties need agreement over shared data without a central controller. DLT means each node keeps its own journal, and the blockchain is the unified shared view that emerges when those journals synchronize through consensus.